ЛЕТО С AIRI 2026

When does DoRA help?
Geometry, budgets, and confirmatory few-shot adaptation

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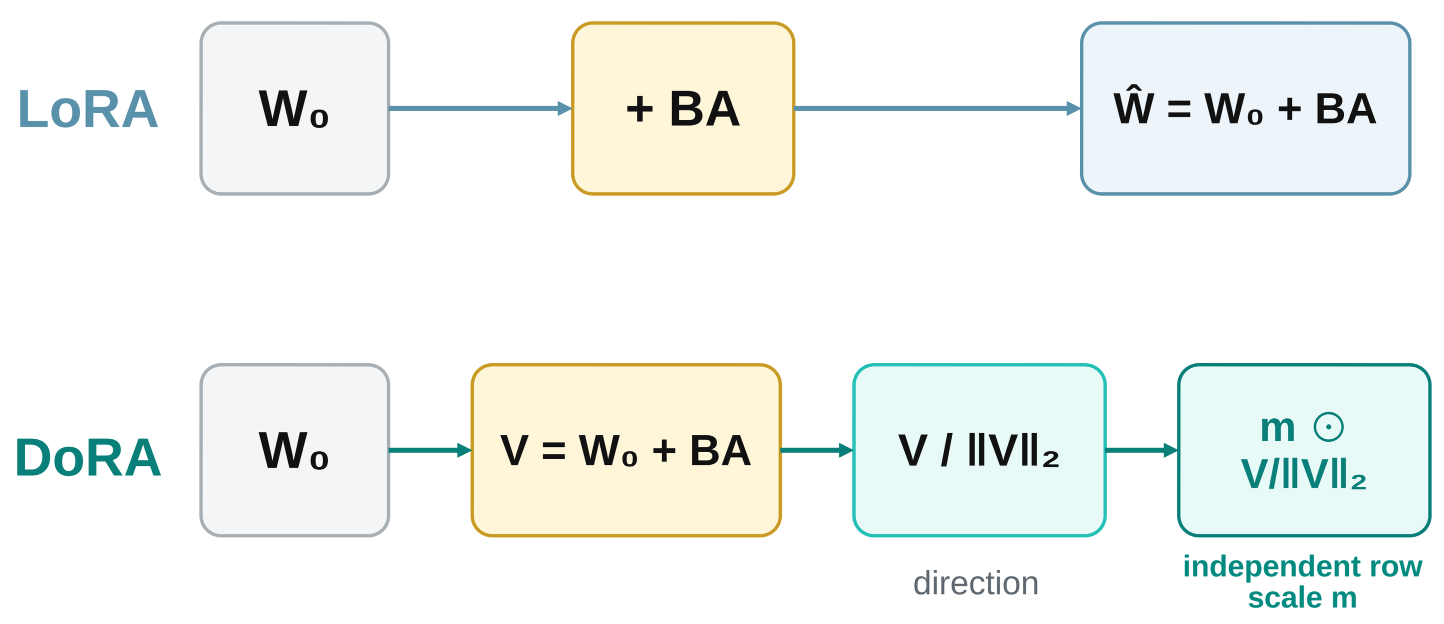
Why separate magnitude from direction?

LoRA adds a low-rank update BA, coupling row direction and row norm. DoRA writes $W = m \odot (W_0 + BA) / \|W_0 + BA\|_2$, so BA controls direction while m controls row scale.

Research question: does this extra degree of freedom improve target adaptation after fair selection, budget controls, and negative controls?



Same rank r; DoRA adds d_out trainable magnitude parameters.

Figure 1. LoRA couples scale and direction; DoRA models row scale explicitly.

Frozen protocol before confirmatory testing

Pilot only (validation): learning rate, early stopping, and rank allocation.
Confirmatory: 20 paired MLP seeds + 10 paired CNN seeds; 3 shifts; 9 pre-declared primary comparisons with Holm correction.
Baselines: frozen, magnitude-only, LoRA, LoRA+, matched-budget LoRA, budgeted DoRA, and full fine-tuning.
Robustness: 4 nested data regimes; 10 synthetic targets × 5 initializations.
1,960 saved result rows; 1,840 independent training / optimization jobs.

Negative controls prevent a victory-only story

Contrast (MLP): magnitude-only 96.90% > DoRA 95.18%.
Rotation (MLP): matched-budget LoRA = DoRA = 93.89%.
Mixed (MLP): matching the parameter budget reduces the DoRA gap to +0.53 pp; its CI crosses zero.

Scope: Digits, one fixed pretrained base per architecture, and one target split already seen during exploratory work. This is a controlled mechanism study, not an LLM-scale benchmark.

CONFIRMATORY MIXED-SHIFT EFFECT

+1.06 pp MLP · +0.92 pp CNN

15/20 MLP seeds and **8/10** CNN seeds favored DoRA. The unadjusted 95% CI excludes zero for CNN, not MLP; **Holm-adjusted** comparison families remain inconclusive.

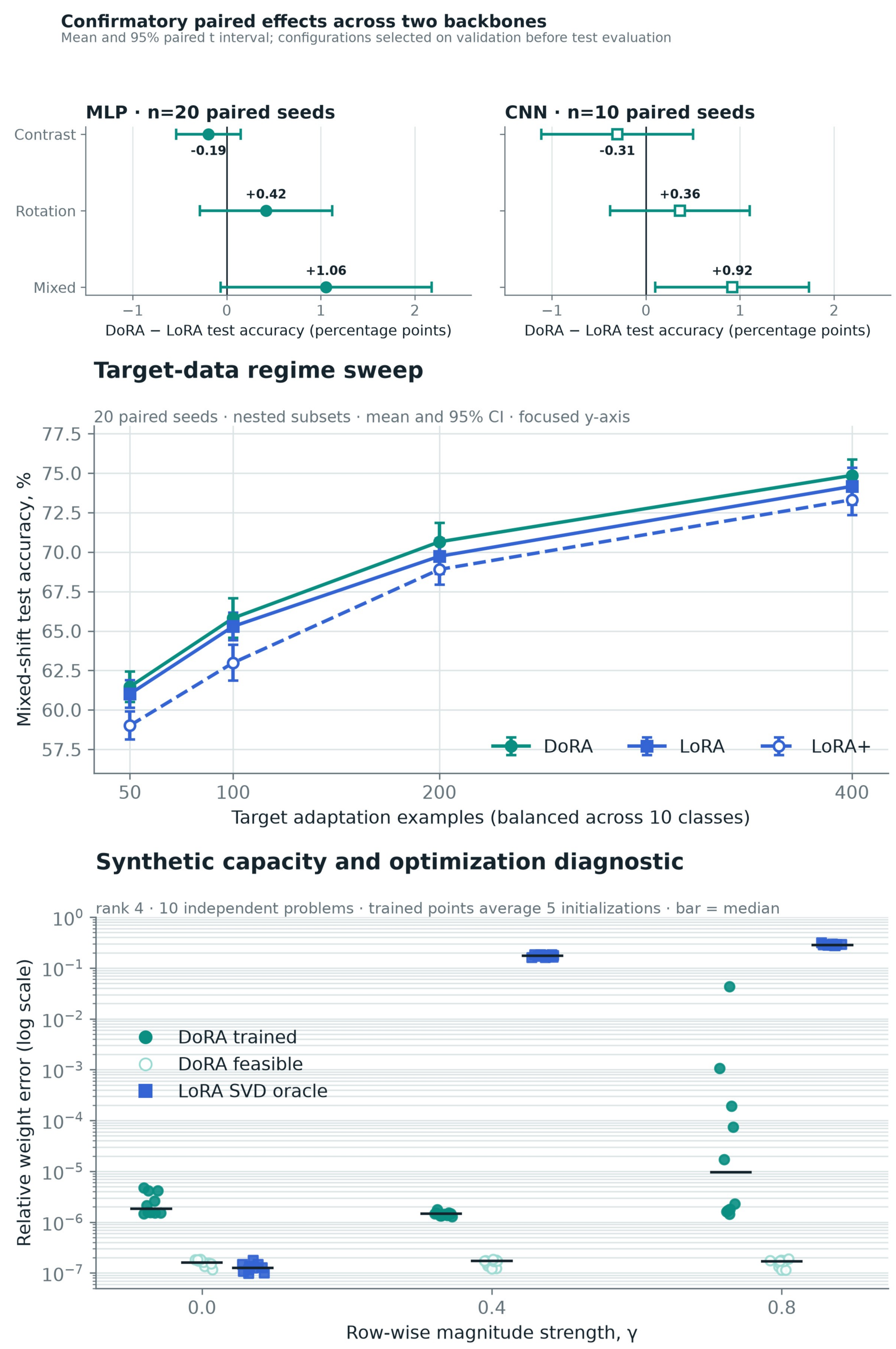


Figure 2. Replication → data regime → capacity / optimization diagnostic.

Target-test evaluation followed the frozen validation-selected extension protocol. Error bars are paired t intervals; exact CIs, p-values, and rank budgets are in the repository.

What survived the controls

- **Mixed-shift DoRA** reaches 75.22% on MLP and 80.53% on CNN - near full FT (75.17%, 80.67%) with only ~12-15% as many trainable parameters.
- **Matched-budget LoRA** narrows the MLP gap to +0.53 pp; budgeted DoRA still gains +0.82 pp over uniform LoRA.
- **At γ=.8**, DoRA is representationally exact; training converges below 10^{-3} in 44/50 runs. LoRA's best possible mean error is 0.289.
- **Decision rule:** test DoRA when heterogeneous row-norm drift is plausible; retain LoRA and magnitude-only baselines.

[1] Hu et al. LoRA. ICLR 2022. [2] Liu et al. DoRA. ICML 2024.

[3] Hayou et al. LoRA+. ICML 2024. Statistical protocol and raw CSV: GitHub.

Code, executed notebook, full report, raw results, and reproduction instructions → QR.

CODE / DATA / REPORT